
Performance Testing Assignment

Purpose

This project is designed to demonstrate your understanding of performance testing by:

- Utilizing appropriate performance testing tools.
- Developing effective test cases.
- Critically analyzing the chosen tool's effectiveness in a real-world scenario.

Objectives

Learners will be able to:

- **Develop a Performance Testing Strategy:** Design tests for an algorithm based on different load levels.
- **Implement Performance Tests:** Create and execute tests using a performance testing framework of your choice.
- **Analyze Test Outcomes:** Evaluate the performance data and determine the effectiveness of your tests in identifying code bottlenecks.

Technology Requirements

- Performance testing framework (of choice)
- IDE (of choice)

Project Description

You will receive a sample code simulating the features of a typical application. Your task is to test this code under varying load conditions using a performance testing framework. The project involves designing test cases, executing the tests, and analyzing the results to identify performance bottlenecks.

Directions

Task 1 - Identification of Framework

You are asked to research and identify a load-testing framework that you will use to test the code given to you using varying levels of loads. In your report, describe the tool you have selected and make a reference to at least one training module that you completed to learn about the features of the tool.

Task 2 - Test Case Design

You are given a sample code in which you will create load tests. The code given simulates the real-world features of a typical application. You should follow the load testing strategy, which is focused on evaluating the performance of the code under varying three (3) levels of load:

- Light load - testing the code with 100 users
- Medium load - testing the code with 1,000 users,
- Large Load - testing the code with 10,000 users

You are asked to develop test cases that simulate these three loads using the framework you have identified in Task 1.

Your report should include screenshots of the test cases (scripts, code).

Task 3 - Monitoring and Development of Execution Report

You are asked to monitor the performance of the code under load test and produce an execution report. The report should detail the specific metrics you observed during the test and explain why these metrics were chosen. Depending on the framework you selected in Task 1, the outcome of the test execution can be reported in many different ways. Your report should include screenshots of all the report(s) generated showing the following information:

- Requests per second (RPS)
- Response time
- Failures in real-time

Task 4 - Analysis

Based on the outcome of the test results, you are asked to analyze the outcome of your load tests. Your report should describe the bottlenecks observed in the code and explain why there is a bottleneck.

Observe the outcome of 3 tests you completed using 3 different loads and discuss which tests were able to identify bottlenecks in the code.

In your analysis, discuss the Failures, Response Times and Request Per Second (RPS) being reported for each load you tested and discuss whether the load tested out is showing any failures in the code or not. In case a failure is observed, discuss when a failure is observed and where in the code failures are being reported. Discuss why you think the failure is being observed.

Task 5 - Improvement of test strategy

In this section, you are asked to update the load testing strategy given to you in Task 2. The loading testing strategy should have decision regarding the type of load needed to cause stress and identify failures in the given code.

Based on your analysis in Task 4, devise a **strategy** to stress the code so that you can identify bottlenecks in the code.

Your report should describe your strategy and explain why your strategy can identify additional bottlenecks not identified before using the load specs provided in Task 2.

Implement your strategy by modifying the load and show the outcome of the strategy. Include a report/evidence showing the outcome of the implementation of this strategy. Your report should discuss whether the recommended strategy identifies additional bottlenecks or not.

Submission Directions for Project Deliverables

You are given an unlimited number of attempts to submit your best work. The number of attempts is given to anticipate any submission errors you may have regarding properly submitting your best work within the deadline (e.g., accidentally submitting the wrong paper). It is not meant for you to receive multiple rounds of feedback and then one (1) final submission. Only your most recent submission will be assessed.

You must submit your project deliverable in its submission space in the course. Learners may not email or use other means to submit any assignment or project for review, including feedback, and grading.

The project includes one (1) deliverable:

- **Report:** Submit your PDF report to the submission space in this course. Title your file as “yourlastname_firstname_CSE 565_LoadTestingProject”.

Your report must contain:

1. **Identification of the framework-** The report should describe the framework and link to the training module used.
2. **Test case design** - The report should describe the test cases developed and include a screenshot(s) of the scripts/code used for varying loads
3. **Test execution report** - The report should describe the outcome of the load tests and include a screenshot(s) of the report generated.
4. **Analysis of test execution report** - The report should describe the analysis of the test report and pinpoint bottleneck(s) in the code.
5. **Improvement of test strategy** - The report should assess the test strategy used, discuss the weaknesses or strengths of the strategy, and recommend additional improvements in the load testing strategy so that additional bottlenecks can be observed in the code.

Evaluation

Your assignment will be evaluated based on the criteria (**worth 100 points total**):

1. **Identification of the framework (5 points)**
2. **Test case design (25 points)**
3. **Monitoring and test execution report (20 points)**
4. **Analysis of test execution report (20 points)**
5. **Improvement of test strategy (25 points)**
6. **Format of paper (5 points)**